

DG4NLP Natural Language Processing

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Week 2: Coursework - Section 2

In this section, you will apply one or more machine learning methods to the dataset prepared in Section 1. The aim is to develop models that can classify scientific papers by category and produce meaningful summaries of their content.

You are expected to take an experimental approach: explore different methods, compare their performance, and clearly document your findings.

Task Overview

You will build a complete NLP pipeline that includes:

- Feature extraction from text data
- Training and evaluating classification models
- Applying a summarisation technique
- Analysing and documenting your results

Steps:

Step 1: Feature Representation (Essential for classical Machine Learning methods)

Transform the textual data into numerical representations suitable for machine learning. You may use standard techniques such as TF-IDF or explore more advanced representations if desired.

You are encouraged to experiment with different input configurations and justify your choices.

Step 2: Text Classification

Develop at least one machine learning model to classify papers into their main category. Suitable approaches include (but are not limited to) Logistic Regression, Naïve Bayes, or Support Vector Machines.

- Split the data into training and testing sets
- Train your model(s)
- Evaluate performance using appropriate metrics

You are encouraged to experiment with multiple Machine Learning models (Classic Machine Learning, Deep Learning, Transformers) or parameter settings and compare their performance.

Step 3: Summarisation

Implement a method to generate summaries of scientific abstracts. This can be a simple extractive approach or a more advanced method if you wish.

Provide examples of generated summaries and briefly comment on their quality.

Step 4: Experimentation and Innovation

This section is intentionally open-ended. You are encouraged to:

- Try different models, features, or preprocessing techniques
- Compare alternative approaches
- Explore improvements beyond the basic implementation

Any additional experimentation should be clearly motivated and explained.

Step 5: Documentation of Experiments

A key part of this section is documenting your work clearly. Your notebook should include:

- A description of the methods you implemented
- Details of any experiments you conducted
- Results of each experiment
- Clear comparison between different approaches
- Brief discussion of what worked well and what did not

Your work should demonstrate not only correct implementation, but also critical thinking and reflection on your experimental results.

Coursework - Section 2 outcome:

- Code:
 - Well-structured and readable code
 - Fully commented to explain each step of the workflow
 - Implementation of:
 - At least one classification model
 - At least one summarisation method
 - Evidence of experimentation, including:
 - Trying different approaches
 - Evaluating model performance using appropriate metrics
 - Comparing results between different methods or configurations
- Documentation:
 - Methods
 - A clear explanation of:
 - The feature extraction approach used (if applicable)
 - The classification model(s) implemented
 - The summarisation technique(s) used

- Experiments and Evaluation
 - Description of experiments conducted
 - Evaluation methods and metrics used
 - Comparison of different models or approaches
- Results
 - Presentation of key results in a clear and structured way
 - At least two example outputs, such as:
 - Model predictions for classification
 - Generated summaries for selected papers
- Reflection
 - Brief discussion of what worked well
 - Limitations of your approach
 - Possible improvements or extensions

After completing this task, you will gain practical experience in applying machine learning techniques to real-world NLP tasks, and in systematically evaluating and documenting experimental results.